

Stochastic Dominance and Infinite Social Welfare

Presentation notes

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Abstract

This paper explores problems that arise when we bring a basic tool for ranking prospects under uncertainty — *first-order stochastic dominance* — to bear on the problem of impartial social welfare in an infinite setting (often framed as the problem of *intergenerational equity*). It turns out that stochastic dominance has serious pathologies when it is applied to random variables on standard (σ -additive) probability spaces taking values in the kind of rich infinitary spaces suitable for representing social preferences that respect Pareto (*ex post*). For example, there are welfare-stream-valued random variables X and Y such that X *weakly stochastically dominates* Y , even though Y *strictly pointwise dominates* X . On the other hand, we can recover a well-behaved notion of stochastic dominance if social preferences over prospects are instead represented using *non-Archimedean* probability spaces.

Impartial Social Welfare and Risk

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We're about to decide where to bury some toxic waste, which may have effects on future generations on into the indefinitely distant future. We've learned from Ramsey (1928) that temporal discounting is "ethically indefensible". So we're *impartial*.

This we means we face a bunch of impossibility results. But I think we can navigate those.

We also don't *know* how our toxic waste is going to affect future people. There are risks involved. And impartial infinite social welfare faces a whole *new* battery of problems in the context of risk. For example, we face technical difficulties making any sense of *expectations* in the infinite setting, as well as even more fundamental clashes. Exploration of these problems is an active research area, but still at an early stage.¹

One of our most basic tools for evaluating risky prospects is (*first-order*) *stochastic dominance*.² But it turns out that in the setting of infinite social

¹Some important work on this problem: Vallentyne and Kagan (1997); Fleurbaey and Michel (2003); Lauwers and Vallentyne (2004); Pivato (2014); Zhou (1997); Vallentyne and Kagan (1997); Lauwers and Vallentyne (2004); Arntzenius (2014); Nover and Hájek (2004); Fine (2008); Seidenfeld, Schervish, and Kadane (2009); Colyvan and Hájek (2016); Goodsell (2021); Askill (2018); Wilkinson (2022); Pivato (2014); McCarthy, Mikkola, and Thomas (2019); Russell and Isaacs (2021); Hong and Russell (2025); Nebel (2025).

²For philosophical and technical discussion, see Easwaran (2014a); Bader (2018); Tarsney

welfare, even our most basic tools don't work very well. Stochastic dominance turns out to have some bad pathologies. I'll show what goes wrong. Then I'll propose a response to these problems based on non-Archimedean probability theory.

Stochastic dominance

What is stochastic dominance?

(Don't worry about "Polish space" or "Radon measure" if you're not familiar. They're just certain nice topological spaces and measures, and they probably include all of the natural examples that first come to mind.)

The standard set-up.

Let $(\mathcal{X}, \geq)$ be a Polish space equipped with a **closed preorder**.

Let $\Delta\mathcal{X}$ be the set of all Radon probability measures on $\mathcal{X}$.

Theorem 1 (Strassen 1965, Edwards 1978). *For any $P, Q \in \Delta\mathcal{X}$, the following are equivalent:*

(a) (**Cumulative dominance**)

$$P(U) \geq Q(U) \text{ for every upward-closed } U \subseteq \mathcal{X};$$

(b) (**Realization**) *There exist $\mathcal{X}$ -valued random variables X, Y (on some probability space) with distributions P, Q respectively, such that*

$$X \geq Y \text{ pointwise};$$

(c) (**Coupling**) *There exists $R \in \Delta(\mathcal{X} \times \mathcal{X})$ with marginals P and Q such that*

$$R(\geq) = 1;$$

(d) (**Expected utility**) *For every bounded monotone function $u : \mathcal{X} \rightarrow \mathbb{R}$,*

$$\mathbb{E}_P u \geq \mathbb{E}_Q u$$

(forthcoming); Tarsney (manuscript); Pomatto, Strack, and Tamuz (2020); Russell (forthcoming).

In this case, P **weakly (first-order) stochastically dominates** Q :

$$P \geq_{\text{s.d.}} Q.$$

(For helpful overview see Fritz 2024, sec. 4.)

Condition (a) corresponds to the most familiar definition of stochastic dominance. In the case where $\mathcal{X} = \mathbb{R}$ we can equivalently put it more simply:

$$P[x, \infty) \geq Q[x, \infty) \quad \text{for each } x \in \mathbb{R}$$

In other words, we have *dominance in cdf*.

Condition (b) is what gives stochastic dominance its name: P and Q are the *stochastic profiles* of variables standing in the relation of (statewise) *dominance*.

There is also a corresponding notion of *strong* stochastic dominance.

Theorem 2. For $P, Q \in \Delta\mathcal{X}$, the following are equivalent:

(a) (**Strict part**) $P \geq_{\text{s.d.}} Q$ and $Q \not\geq_{\text{s.d.}} P$,

(b) (**Cumulative dominance**)

$$\begin{aligned} P(U) &\geq Q(U) \text{ for every upward-closed } U \subseteq \mathcal{X}, \text{ and} \\ P(U) &> Q(U) \text{ for some upward-closed } U \subseteq \mathcal{X}. \end{aligned}$$

(c) (**Realization**) There exist $\mathcal{X}$ -valued random variables X, Y with distributions P, Q , respectively, such that

$$\begin{aligned} X &\geq Y \text{ pointwise, and} \\ \mathbb{P}(X > Y) &> 0 \end{aligned}$$

(d) (**Coupling**) There exists $R \in \Delta(\mathcal{X} \times \mathcal{X})$ with marginals P and Q such that

$$\begin{aligned} R(\geq) &= 1, \\ R(>) &> 0 \end{aligned}$$

(e) (**Expected utility**) For every bounded strictly monotone $u : \mathcal{X} \rightarrow$

$\mathbb{R}$,

$$\mathbb{E}_P u > \mathbb{E}_Q u.$$

In this case, we say P **strongly stochastically dominates** Q , written $P \succ_{\text{s.d.}} Q$.

So this is great. As long as our measures and orders are neat and tidy, so is stochastic dominance. It naturally extends the idea of *statewise* dominance from *random variables* to *probability distributions*. In the terms of decision theory — we get a dominance concept that doesn't just make sense for *Savage-style acts*, but also for *von Neumann and Morgenstern-style lotteries*.

This makes the following a very natural axiom for social preference:

Stochastic Dominance Axiom. If a social prospect A (weakly/strongly) stochastically dominates a social prospect B , then $(A \succsim B / A \succ B)$.

Infinite social choice (without risk)

Now we come back to social choice. Some setup.

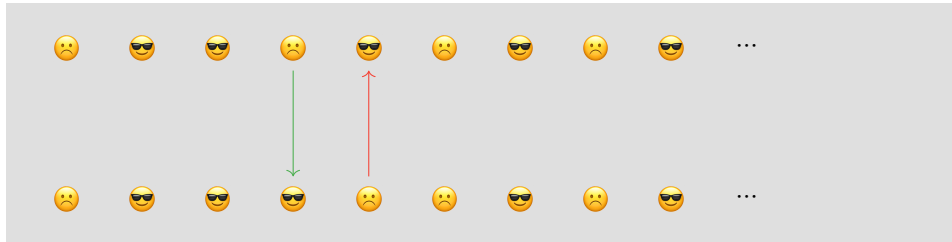
- Let $\mathcal{I}$ be an infinite population.
- Let $(\mathcal{X}, \geq)$ be a preordered set of individual welfare levels^a
- **Social outcomes** are elements of $\mathcal{X}^{\mathcal{I}}$.
- A **social welfare relation** is a preorder on $\mathcal{X}^{\mathcal{I}}$.

^a... containing at least two elements $x > y$, where $\mathcal{X}$ is a Hausdorff topological space.

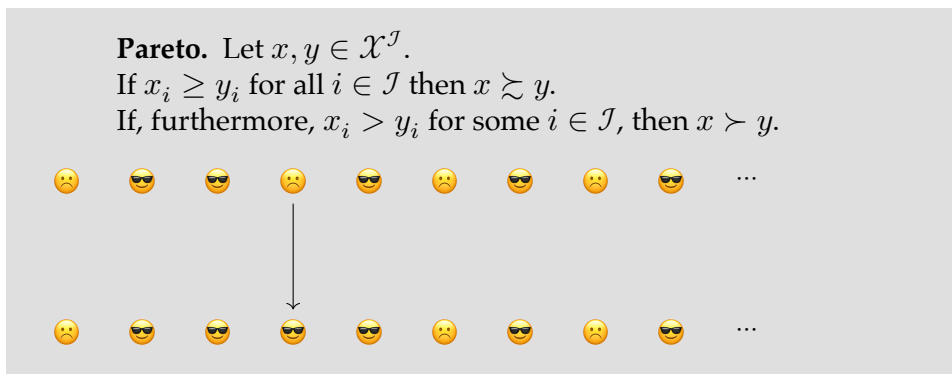
Here are some standard axioms.

Harming one individual is just as bad as harming any other individual, regardless of how far into the future they may live.

Impartiality (Finite Anonymity). Let $x, y \in \mathcal{X}^{\mathcal{I}}$. If $x_i = y_i$ for all $i \in \mathcal{I}$ other than j, k , and $x_j = y_k$ and $x_k = y_j$, then $x \sim y$.



We're also *Paretian*: we prefer options that certainly benefit some individuals, if they certainly don't make anyone worse off.



Now that we're in this predicament, we face some familiar challenges.

Impossibility results!

- No impartial Paretian social preferences have a real-valued utility representation (Diamond 1967; Basu and Mitra 2003).
- No impartial Paretian social preferences are consistent with *strong anonymity* (indifference between arbitrary permutations) (see Lauwers 1997).
- No *complete* impartial Paretian social preferences are consistent with *permutation invariance* (Askill 2018; Goodman 2025).
- The existence of complete impartial Paretian social preferences cannot be constructively proved (Lauwers 2010).

Nonetheless, these problems are not insurmountable.

(e.g. Vallentyne and Kagan 1997; Fleurbaey and Michel 2003; Lauwers and Vallentyne 2004; Pivato 2014).

Here is a central, simple example of an impartial Paretian social welfare relation.

Example 1 (Counting order). • Let $\mathcal{X} = 2 := \{0, 1\}$.

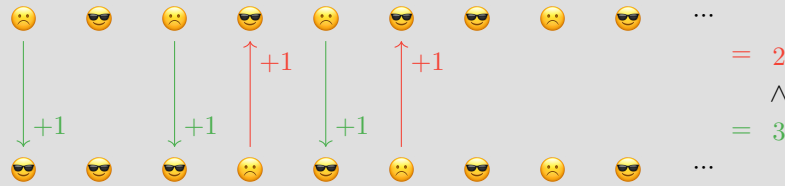
- For $x, y \in \mathcal{X}^{\mathcal{J}}$, let

$$d(x, y) := \#\{i \in \mathcal{J} \mid x_i = 1 \text{ and } y_i = 0\}$$

- For $x, y \in \mathcal{X}^{\mathcal{J}}$, let

$$x \geq^{\#} y \text{ iff } d(y, x) \text{ is finite and } d(x, y) \geq d(y, x)$$

(including the case where $d(x, y)$ is infinite).



Fact 1. Any impartial Paretian social welfare relation embeds the counting order.

This means that, for the purpose of *negative* results, we can often just focus on the simple case of the counting order on 2^{ω} .

Stochastic dominance and infinite social choice at the same time (uh oh)

Now we run into our first technical problem. Recall that the equivalences for stochastic dominance were stated for a *closed* preorder (on a Polish space).

Fact 2. Let 2^{ω} be equipped with the product topology.

The counting order on 2^{ω} is **not closed**.

More generally, no impartial Pareto order on $\mathcal{X}^{\mathcal{J}}$ is closed (in the product topology).

Proof. The sequences $x_k = (0, \dots, 0, 1, 1, \dots)$ and $y_k = (1, \dots, 1, 0, 0, \dots)$ satisfy $x_k >_{\#} y_k$ for each k . But in the product topology,

$$\lim_k x_k = (0, 0, 0, \dots)$$

$$\lim_k y_k = (1, 1, 1, \dots)$$

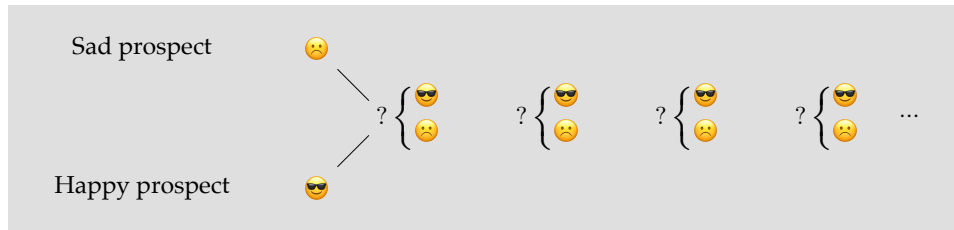
and we have $(0, 0, 0, \dots) \not\geq_{\#} (1, 1, 1, \dots)$. So the order is not closed.

Since any impartial Pareto order embeds the counting order, similar reasoning applies. $\square$

This is crucial: the topological condition that keeps stochastic dominance orderly in general settings does not hold for infinite social welfare orderings. In fact, things break down even in very simple and natural cases.

Consider the **infinite random world**. Imagine a world containing countably many individuals. Each individual is either happy or sad, and each individual's welfare is determined by the toss of an independent fair coin. In this case the distribution over social outcomes is simply the coin flip distribution on 2^{ω} .

We can modify this case by making one person *certainly happy*. Or we can modify it a different way, making one person *certainly sad*. Intuitively, the first option is socially preferable.





Example 2. • Let B be the uniform measure on $\{0, 1\}$.

- The **standard coin flip space** is the infinite product space $\Omega = 2^\omega$ with the product measure B^ω .
- Consider random variables $X, Y : \Omega \rightarrow 2^\omega$, respectively, such that

$$X(x_2, x_3, \dots) := (1, x_2, x_3, \dots)$$

$$Y(x_2, x_3, \dots) := (0, x_2, x_3, \dots)$$

- X is a statewise Pareto improvement over Y .
- X and Y have respective distributions

$$P := \delta(1) \otimes B^\omega$$

$$Q := \delta(0) \otimes B^\omega$$

(These are Radon measures on 2^ω with the counting order.)

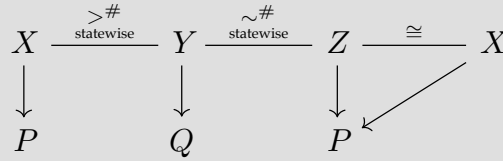
Here is the central fact.

Fact 3. *There exists a random variable $Z : \Omega \rightarrow 2^\omega$ such that*

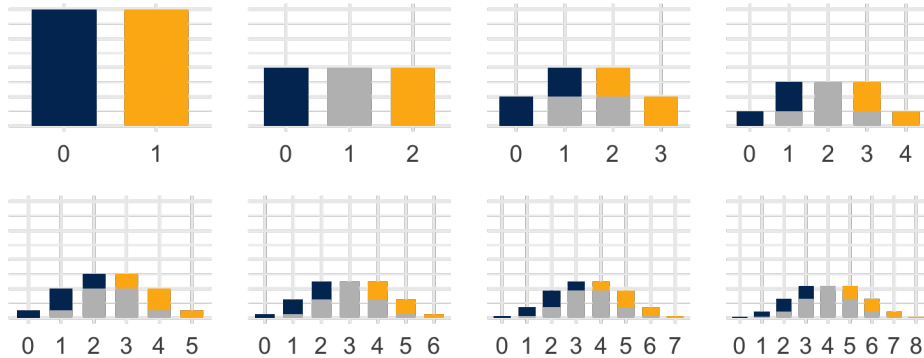
- $Z \sim^\# Y$ statewise, and also
- *There exists a probability space isomorphism $\pi : \Omega \rightarrow \Omega$ such that*

$$Z = X \circ \pi \quad \text{almost everywhere};$$

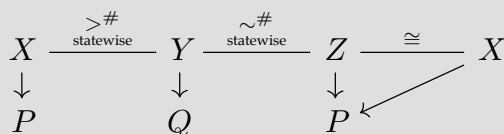
- *Thus X and Z have the very same distribution.*



Proof idea. For any n , let $f_n(x_1, x_2, \dots)$ be the number of 1's that occur in the first n places in the sequence. Here are the distributions of $f_n(X)$ and $f_n(Y)$ (??). These are two binomial distributions, offset by one place. The key thing is that they get closer and closer together. For each n , the gray region represents a pair of events E_n and F_n , related by a probability preserving permutation of states π_n , such that X on E_n and Y on F_n are equivalent up to a finite permutation of individuals. And in the limit, the events E_n has (standard) probability one. So the random variable $Z = X \circ \pi$ is statewise equivalent to Y and probabilistically equivalent to X . $\square$



This means our several different formulations of strong stochastic dominance come apart, so we'll have to carefully distinguish them.



Consequences.

1. Q **weakly stochastically dominates** P
 - (in all four senses).
2. P **strongly stochastically dominates itself**
 - in the realization and coupling senses, and
 - in the expected utility sense, vacuously
 - (but not in the strict part or cumulative dominance senses).
3. Q **strongly stochastically dominates** P
 - in the same three senses.
4. Neither part of the Stochastic Dominance axiom is plausible.

Thus *neither* part of the proposed Stochastic Dominance axiom is plausible in this setting. Weak stochastic dominance does not suffice for weak betterness. (Clearly Q is not as good as P .) Likewise, strong stochastic dominance (in the realization, coupling, or EU sense) does not suffice for strict betterness. (Clearly P is not better than P .) Nor do these conditions suffice even for *weak* betterness (since Q is not as good as P).

Moreover, this situation is pervasive. Appending an infinite background population of fair coin flips to *any* finite prospects completely washes out any differences in their probability distributions over outcomes, up to social indifference. Thus a prospect that is certain to produce a strictly better social outcome has essentially the same distribution as one that produces a strictly worse outcome.

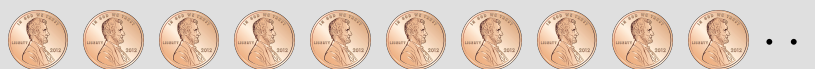
We are used to thinking that, at least in contexts where all the probabilities are

known, it is harmless to switch from the problem of ordering *social prospects* to the problem of ordering *probability distributions*. (That is, given a probability measure on states, we can normally move harmlessly from a Savage-style representation to a von Neumann and Morgenstern-style representation.) We might lose a bit of fine detail, corresponding to differences on events with probability zero, but normally these fine details are irrelevant to ordering prospects. But it turns out that in the setting of infinite social choice, *all* finite differences between acts disappear through the cracks of measure zero events. Classical probability is too coarse to capture the relevant distinctions.

$$X = Z \circ \pi \quad \text{almost everywhere}$$



Non-Archimedean probability



$$P(\text{all heads}) > 0$$

My approach builds on work by Benci, Horsten, and Wenmackers (2013); Benci, Horsten, and Wenmackers (2018); Pivato (2014).

This suggests a way forward: we should use representations of probability that keep track of the differences between different sets that all classically have probability zero. In short, we can turn to *non-standard analysis*, and specifically *non-Archimedean probabilities*. This turns out to be a promising direction. Many questions about this approach remain open, but here I will sketch some first steps in this direction.³

I don't have very much time, so I'm just going to gesture in the direction this is headed.

Instead of representing probabilities with *real* numbers, we will choose abstract quantitative representations with the right kind of structure to make sense of probability. Recall the usual Kolmogorov axioms:

1. $P(\Omega) = 1$.
2. $P(A) \geq 0$.
3. $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$.

Probability measures make sense for any **partially ordered commutative ring (poc-ring)** $\mathcal{R}$: a set with well-behaved

- ordering,
- addition (with 0), and
- multiplication (with 1).

For these to make sense, we need probabilities to have some kind of *ordering*, and some kind of *addition*. Furthermore, the probability 1 gets special treatment. Since 1 is the multiplicative identity, this suggests that we also need some kind of *multiplication* (which is born out in the theory of *conditional probability*). The natural kind of structure that combines these features is a *partially ordered commutative ring*, or *poc-ring* for short.⁴

³For the key insights here, I draw heavily on Benci, Horsten, and Wenmackers (2013); Benci, Horsten, and Wenmackers (2018); Pivato (2014). The first technical innovation in this section is to move from *totally ordered fields* to *partially ordered commutative rings* — thus providing a fully *constructive* approach to non-Archimedean probabilities. The second is a novel representation theorem for uniform regular NA-probabilities, and some of the points about the coin flip measure.

⁴In this way the approach is different from a more familiar application of non-standard

For any poc-ring $\mathcal{R}$, we can define $\mathcal{R}$ -valued **probability measures** in the standard way, just as above. But it is useful to consider rings and measures that have a bit more structure.

More?

Standard probabilities also have a **continuity** property.

$$P(A) = \lim_{n \rightarrow \infty} P(A_n) \quad \text{whenever } A_n \uparrow A$$

We don't want that, but we do want something analogous.

Standard probability extends the axioms just stated with a *continuity* axiom, telling us about the *limits* of probabilities of increasing sequences of sets. (Countable additivity is equivalent to the combination of finite additivity with such a continuity principle.) This continuity axiom is part of the problem when it comes to infinite social choice, but we don't want to *entirely* dispense with the idea. In our more general probability rings, the standard topological notion of a limit will not be very useful, but we can replace it with a more general idea that will let us bridge from the finite to the infinite (Benci, Horsten, and Wenmackers 2018).

For any indexing set I , the set of $I \rightarrow \mathbb{Q}$ functions naturally inherits the structure of a poc-ring, where the order and arithmetic operations are defined pointwise. Furthermore, (for I non-empty) there is a natural embedding of $\mathbb{Q}$ in this ring, sending each rational number to its corresponding constant function. (We can do this with real numbers rather than rationals, but that actually turns out to be overkill.)

Definition 1. An (I) -**limiting ring** is a poc-ring $\mathcal{R}$ equipped with an operator

$$\lim^{\mathcal{R}} : \mathbb{Q}^I \rightarrow \mathcal{R}$$

analysis to probability theory, using *hyperreals* (Robinson 1966; Bernstein and Wattenberg 1969; Loeb 1975; for philosophical discussion see Skyrms 1980, 1995; for a more critical response see Easwaran 2014b). Hyperreals form a *totally ordered field*. In contrast, our “probabilities” are not totally ordered, and they do not allow for arbitrary non-zero division. These turn out to be features, not bugs: they make it possible to *constructively define* non-Archimedean probabilities, and to preserve attractive symmetries.

Technically, we could get by with a *semiring* rather than a ring — we have no need for *negative* probabilities. But we won't really have use for that additional generality.

which is a monotone ring homomorphism and one-to-one on constant functions.^a

^aThis approach is inspired by Benci, Horsten, and Wenmackers (2013); Benci, Horsten, and Wenmackers (2018); Pivato (2014).

We will normally omit the superscript $\mathcal{R}$; but this limit operator should not be confused with the standard topological limit.

If I is non-empty, then any I -limiting ring $\mathcal{R}$ contains an isomorphic copy of $\mathbb{Q}$, which we will identify with $\mathbb{R}$ itself. Typically, $\mathcal{R}$ properly *extends* $\mathbb{Q}$, and in fact is non-Archimedean.

(A convenient way to construct I -limiting rings is as *filter powers* $\mathbb{Q}^{\mathcal{F}}$, where $\mathcal{F}$ is some filter on I . Every I -limiting ring contains an isomorphic copy of some such $\mathbb{Q}^{\mathcal{F}}$.)

It turns out that the limit operator for *real-valued* functions uniquely determines a very general kind of limit operator for *any* kind of function on I .

Fact 4. *Let $\mathcal{R}$ be a limiting ring. Let A be any structure. There exists a (unique up to isomorphism) structure $A^{\mathcal{R}}$ together with a **limit operator***

$$\lim^{\mathcal{R}} : A^I \rightarrow A^{\mathcal{R}}$$

such that $\lim^{\mathcal{R}}$ is a surjective homomorphism, and^a

$$\lim_i^{\mathcal{R}} a_i = \lim_i^{\mathcal{R}} b_i \quad \text{iff} \quad \lim_i^{\mathcal{R}} 1_{[a=b]} = 1 \quad \text{for each } a, b \in A^I$$

(Here $1_{[a=b]}$ is the indicator function for the set $\{i \in I \mid a_i = b_i\}$.)

We call $A^{\mathcal{R}}$ the **$\mathcal{R}$ -power of A** .

^aThe right-hand side is the original limit operator from $\mathcal{R}$ for *rational-valued functions*.

Note that when A has nice structure, the $\mathcal{R}$ -power $A^{\mathcal{R}}$ naturally inherits lots of it.⁵

⁵In fact, $\mathcal{R}$ -powers of a structure are just a repackaging of **filter powers**, also called **reduced powers**. In general, a filter power satisfies the same *Horn theory* as its underlying structure.

Now the core idea is that a general distribution is, in a sense, a **limit of finite-support distributions**.

Definition 2. Let $\Delta_0^{\mathbb{Q}}\mathcal{X}$ be the set of all rational probability distributions on $\mathcal{X}$ with finite support.

For any limiting ring $\mathcal{R}$, let

$$\Delta^{\mathcal{R}}\mathcal{X} := (\Delta_0^{\mathbb{Q}}\mathcal{X})^{\mathcal{R}}.$$

The elements of $\Delta^{\mathcal{R}}\mathcal{X}$ are called **$\mathcal{R}$ -distributions**.

Then $\mathcal{R}$ -distributions straightforwardly inherit important structure from finite-support distributions.

Definition 3. Let $P = \lim_i^{\mathcal{R}} P_i$ be an $\mathcal{R}$ -distribution on a set $\mathcal{X}$.

(a) Define an **$\mathcal{R}$ -valued probability measure** $[P]$:

$$[P](A) = \lim_i^{\mathcal{R}} P_i(A) \in \mathcal{R} \quad \text{for each } A \subseteq \mathcal{X}$$

(b) Let V be any partially ordered (rational) vector space.

Define an **expectation operator** $\mathbb{E}_P$:

$$\mathbb{E}_P u = \lim_i^{\mathcal{R}} \mathbb{E}_{P_i} u \in V^{\mathcal{R}} \quad \text{for each } u : \mathcal{X} \rightarrow V$$

for any p.o. (rational) vector space V .

(c) For any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, define the **pushforward** distribution.

$$f * P := \lim_i^{\mathcal{R}} (P_i \circ f^{-1})$$

Theorem 3. Let $\mathcal{R}$ be a limiting ring, let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$. The following are equivalent:

(a) (**Realization**.) There exists $R \in \Delta^{\mathcal{R}}(\mathcal{Y})$ and $X, Y : \mathcal{Y} \rightarrow \mathcal{X}$ such

that

$$P = X * R, \quad Q = Y * R, \quad \text{and } X \geq Y \text{ pointwise.}$$

(b) (**Coupling.**) There exists $R \in \Delta^{\mathcal{R}}(\mathcal{X} \times \mathcal{X})$ such that

$$P = \text{proj}_1 * R, \quad Q = \text{proj}_2 * R, \quad \text{and } [R](\geq) = 1.$$

(c) (**Expected vector utility.**) For any monotone vector-valued function $u : \mathcal{X} \rightarrow V$,

$$\mathbb{E}_P u \geq \mathbb{E}_Q u$$

In this case, we say P **weakly stochastically dominates** Q , written $P \geq_{s.d.} Q$.

Theorem 4. Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}} \mathcal{X}$ be $\mathcal{R}$ -distributions. The following are equivalent:

(a) (**Strict part.**) $P \geq_{s.d.} Q$ and $Q \not\geq_{s.d.} P$.

(b) (**Realization.**) There exists $R \in \Delta^{\mathcal{R}}(\mathcal{Y})$ and $X, Y : \mathcal{Y} \rightarrow \mathcal{X}$ such that

$$P = X * R, \quad Q = Y * R, \quad X \geq Y \text{ pointwise, and } [R][X > Y] > 0.$$

(c) (**Coupling.**) There exists $R \in \Delta^{\mathcal{R}}(\mathcal{X} \times \mathcal{X})$ such that

$$P = \text{proj}_1 * R, \quad Q = \text{proj}_2 * R, \quad [R](\geq) = 1, \text{ and } [R][>] > 0.$$

(d) (**Expected vector utility.**) For any strictly monotone vector-valued function $u : \mathcal{X} \rightarrow V$,

$$\mathbb{E}_P u > \mathbb{E}_Q u$$

In this case say that P **strongly stochastically dominates** Q , written $P >_{s.d.} Q$.

What about **cumulative dominance**?

$$[P](U) \geq [Q](U) \quad \text{for each upward-closed } U \subseteq \mathcal{X}$$

$$\begin{array}{ccc} \text{stochastic dominance} & \Rightarrow & \text{cumulative dominance} \\ & \Leftarrow & \end{array}$$

Proposition 1. *Weak stochastic dominance is a preorder on $\Delta^{\mathcal{R}}\mathcal{X}$.*

Fact 5. *Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$ be $\mathcal{R}$ -distributions. If $P \geq_{s.d.} Q$, then for all upward-closed subsets $U \subseteq \mathcal{X}$, $P(U) \geq Q(U)$.*

But the converse does not hold in general.

Fact 6. *There exist (additively generated) distributions $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$ such that P cumulatively dominates Q , but P does not stochastically dominate Q .*

Here is a core example.

Example 3. • The **coin flip symmetry group** $G := \bigoplus_{\omega} \mathbb{Z}_2$ naturally acts on 2^{ω} . (Flip any bit.)

- Define **G -symmetric limiting ring** where the indexing set is $\text{Fin } 2^{\omega}$.

$$\lim_{\mathcal{R}} \#(K \setminus \sigma \cdot K) = 0 \quad \text{for each } \sigma \in G$$

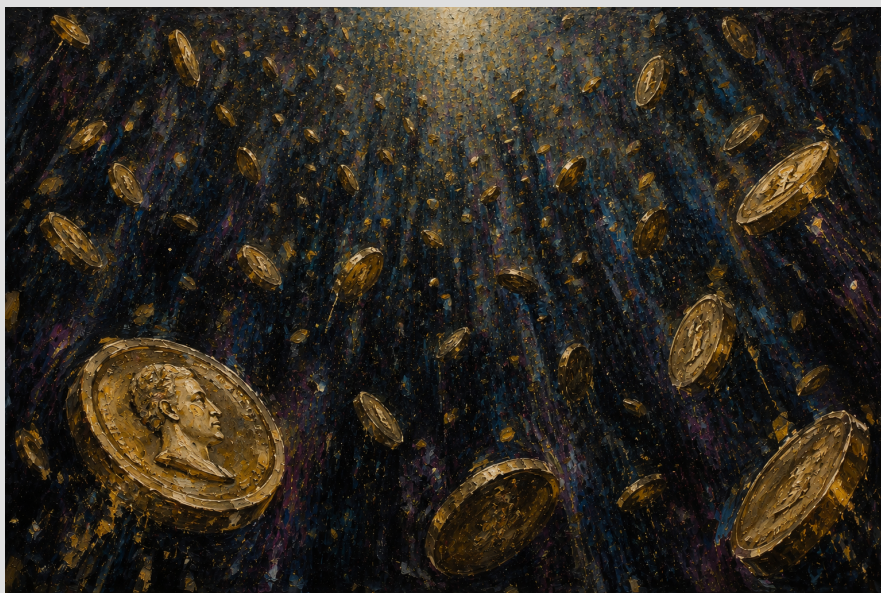
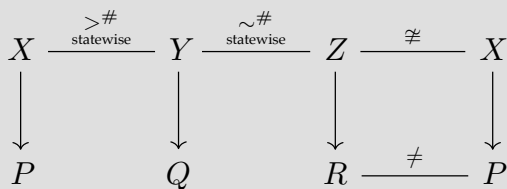
- There is an **initial G -symmetric limiting ring $\mathcal{R}^*$** . Call this the **weak coin flip ring**.
- The **weak coin flip distribution $\mathbb{P}^*$** is the **uniform $\mathcal{R}^*$ -distribution** on 2^{ω} .
- $\mathbb{P}^*$ is symmetric under G :

$$[\mathbb{P}^*](A) = [\mathbb{P}^*](\sigma \cdot A) \quad \text{for each } \sigma \in G, A \subseteq 2^{\omega}$$

Thus $\mathbb{P}^*$ agrees with the standard coin flip measure on all *cylinder sets* — intuitively, these correspond to propositions that are only about the outcomes

of finitely many coin flips.⁶

Now let's return to the notion of stochastic dominance in this non-Archimedean setting.



⁶The weak coin flip order really is *weak*. In fact, outside of cylinder sets it has a lot of indeterminacy. Accordingly, it is not guaranteed to be close to the standard coin flip measure beyond cylinder sets. On the other hand, it *is* guaranteed not to be “determinately far” from the standard measure.

For all Borel sets $A \subseteq 2^\omega$, there is no standard real number $\epsilon > 0$ such that $|\mathbb{P}^*(A) - B^\omega(A)| > \epsilon$ (in $\mathcal{R}$).

But NA-probabilities are only *partially ordered*, this does *not* guarantee that $|\mathbb{P}^*(A) - B^\omega(A)| < \epsilon$, and thus it does not guarantee that $\mathbb{P}^*(A)$ is infinitesimally close to $B^\omega(A)$.

Remember our core example, the infinite random world. There we had two distinct random variables X and Z , where X was a strict Pareto improvement over Z , but X and Z have the very *same* distribution over social outcomes. But remember: we had $X = Z \circ \pi$ *almost everywhere* — they still differed on a set of standard probability zero. The *non-Archimedean* coin flip measure assigns this residue a *positive infinitesimal* probability. So in fact, X and Z have *different* non-Archimedean distributions. In fact, the non-Archimedean probability assigned to each upward-closed set is *infinitesimally higher* for the happy prospect X than it is for the sad prospects Y or Z .

These infinitesimal differences suffice to keep many of the nice properties of stochastic dominance intact — and to keep the plausibility of a *non-Archimedean* form of Stochastic Dominance as an axiom for social choice.

Conclusion



Stochastic dominance is a natural foundation for ranking social prospects under risk. But when it comes to the orderings that naturally arise in the setting of infinite social choice, stochastic dominance falls apart, and ceases to be a good guide to social choice. All finite differences between social acts disappear through the cracks of measure-zero events. The standard probability representation is too coarse to capture the distinctions that matter.

But we don't have to abandon stochastic dominance altogether. Instead, we can move to *non-standard* probability representations, which have the resolution to distinguish distributions that standard probability conflates.

More generally, non-Archimedean probability theory has recently reemerged as an exciting area of active research, and it is a promising setting in which to explore questions about social choice. Lots of important questions remain open, but this seems like a promising start.

Bonus content

What is the precise relationship between $\mathcal{R}$ -distributions and $\mathcal{R}$ -valued probability measures? Are there $\mathcal{R}$ -valued probability measures that do not arise from $\mathcal{R}$ -distributions? (Probably yes.) Can two different $\mathcal{R}$ -distributions give rise to the same $\mathcal{R}$ -valued probability measure?

$\mathcal{R}$ -distributions determines an $\mathcal{R}$ -valued probability measure. Does it contain more information than the measure?

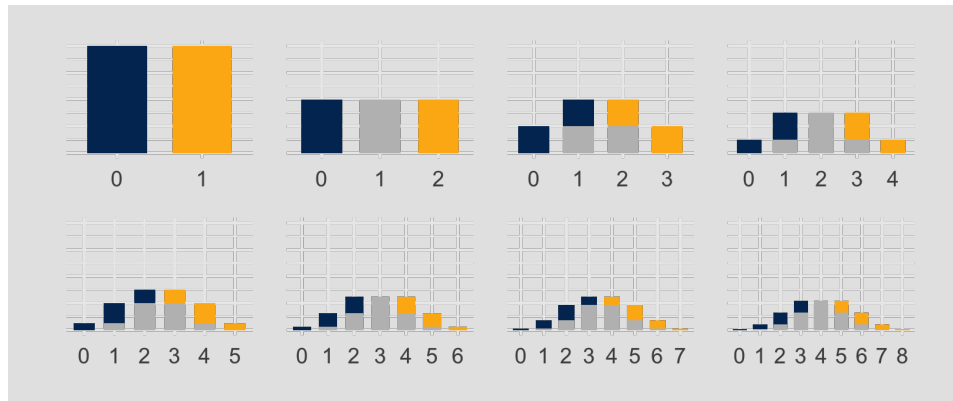
Definition 4. For a $(\text{Fin } \Omega)$ -limiting ring $\mathcal{R}$, an $\mathcal{R}$ -distribution P is **additively generated** iff $P = \lim_K^{\mathcal{R}} P_K$ for some $P \in (\Delta_0^{\mathbb{Q}})^{\text{Fin } \Omega}$ such that

$$P_{J \cup K} = P_J + P_K \quad \text{for each } J, K \in \text{Fin } \Omega \text{ with } J \cap K = \emptyset$$

(The uniform distribution $\mathbb{P}^*$ and all of its pushforward distributions are additively generated.)

Definition 5. Call a $(\text{Fin } \Omega)$ -limiting ring $\mathcal{R}$ **nice** iff for any additively generated $\mathcal{R}$ -distributions P and Q , if $[P] = [Q]$, then $P = Q$.

Theorem 5. *The weak coin flip ring $\mathcal{R}^*$ is nice.*



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